U.PORTO FACULDADE DE CIÊNCIAS UNIVERSIDADE DO PORTO U.PORTO FEUP FACULDADE DE ENGENHARIA UNIVERSIDADE DO PORTO	**Master's Degree in Artificial Intelligence** **Artificial Intelligence and Society**	**2026/2027** **1st Year** **1st Semester**
INSTRUCTOR: Miriam Seoane Santos		

Group Project

The Icarus Dilemma: Progress beyond Responsibility

Theme

This project examines the tension between progress and societal responsibility: when does technological ambition start becoming recklessness? At the pace we are at currently, new capabilities arrive before societies have agreed on their limits, understood their consequences, and decided who should be held accountable when things go wrong. The goal of this group project is to discuss what happens when innovation advances faster than our ability to anticipate, govern, regulate, and respond to consequences.

To that end, each group should **identify a case study** where an emerging AI technology raises the question of whether society has gone (or may be going) too far. The analysis should examine the motivations driving its development, the benefits it promises, the risks and unintended consequences it may create, and the ethical and societal trade-offs involved. Related topics may concern autonomous systems, mass surveillance, human enhancement, synthetic media, biotechnology, military technologies, environmental costs, privacy, human autonomy, inequality, governance, accountability, or other areas where technological capability threatens to advance beyond social responsibility.

Building on the case study analysis, each group is expected to **design and deliver a practical activity** (e.g., a seminar, a workshop, a tutorial, a journalistic piece, a demo, an interactive experience, simulation, or other) that engages the community in critically reflecting on the chosen topic. The activity should move beyond simply presenting information: it should encourage participants to question assumptions and reflect on where the limits of technological progress should lie. In doing so, the group should demonstrate both a strong understanding of the topic and the ability to communicate complex ethical and societal issues in an accessible and engaging way.

The deliverables of this project are:

1. A **final report (blog article)** that documents the case study, its context and analysis, as well as the design of the practical activity.
2. A **hands-on activity** (or demo, interactive experience) presented during the final course sessions, that fosters dialogue on the ethical and societal implications of AI.

Project Overview

- **[required]Oct 05: (Moodle)** Deliver a PDF with information on the planned case study.
- **[optional] Oct 16: (In-class)** Discussion of case study and potential practical activities.
- **[optional] Nov 06: (In-class / Email)** Feedback on project evolution and expectations.
- **[optional] Nov 27: (In-class / Email)** Recommendations for the final presentation and reports.
- **[required] Dec 04: (In-class)** Project Presentations (Part 1 – 5 groups).
- **[required] Dec 11: (In-class)** Project Presentations (Part 2 – 5 groups).
- **[required] Dec 20:** Final Delivery (*after presentation feedback*)

Evaluation

The project will be evaluated regarding the following expected outcomes:

- **Progress and Involvement (3 points):** Active participation, involvement, and progress.
- **Final Report (7 points)**: Depth and quality in the exploration of the case study, the produced insights, critical analysis, and conclusions.
- **Practical Activity (7 points):** Creativity, relevance of the activity, level of engagement.
- **Professionalism and Presentation (3 points)**: Structure and clarity of the final presentation, professionalism and ability to collaborate with others.

All students should participate in the final presentation and the practical activity, and they are graded according to each member's contribution.

Project Phases and Deliverables

A. Project Planning

You should start by finding or selecting a documented case study where AI development has raised ethical or societal concerns. Make sure that your topic contains sufficient information to allow for a meaningful analysis and validate it with the teacher before you start.

Students should submit a short PDF report containing the following:

- Title of the project;
- Description of the project;
- Team Members;
- Team Leader;
- Roles of each team member;
- Preliminary Resources: data, code, videos, blogs, open-source software, other tools;
- Bibliography: books, news, reports, documents, research papers;
- Potential Practical Activities (ideas).

B. Case Study Analysis

Now that you have established your case study, try to provide a comprehensive discussion on the following:

- **Context and Problem Definition:** Describe the context in which the AI system was developed, introduced, or adopted and explain the societal need, ambition or problem it was intended to address. Identify the main actors involved, such as companies, governments, researchers, institutions, users, or affected communities.
- **Technical Aspects:** Discuss the technicalities of how the AI was developed and/or deployed, including relevant factors whenever you can (e.g., dataset quality, algorithmic choices, outcomes, design choices, infrastructure, examples).
- **Outcomes and Implications:** Examine the intended benefits and the actual or potential consequences of the technology. What problems did it solve, and what new problems did it create? Who benefited from its development, and who was exposed to its risks or costs? Consider unintended consequences, possible harms, trade-offs, and whether some risks were overlooked, underestimated, or accepted in the pursuit of progress.
- **Societal Response and Critical Reflection:** Comment on the societal response (if there was any) or provide your own conclusions on the topic. How did society react? Were there legal consequences, public outrage, coverage on the news, political changes? What did the media report on the issue, how did stakeholders (government, public institutions, companies) react?

C. Practical Activity

Following Task B, what have you learned regarding the interaction of society with AI? How do you think it affects public perception of AI systems? What values, ideas, assumptions are at stake? How can we get involved, as citizens, educators, policymakers, and technical specialists to contribute to more responsible practices and a trustworthy use of AI?

This task involves producing a practical contribution on that topic. Look into related projects, read academic papers, articles, and reports to learn more about the topic and how it relates to your case study. Then, propose a practical activity where you can engage with the community. This component is open-ended and can comprise different approaches. Some examples are:

1. Facilitating a seminar, a workshop, or creating a video-series to demystify certain concepts;
2. Conducting street interviews on campus, a survey or a poll, and turn them into a short video or report;
3. Setting up an interactive exhibit (e.g., a demo, a booth, a quizz/game, a multimedia product, a game);
4. Designing an awareness campain;
5. Hosting a debate to re-enact stakeholder perspectives (e.g., developer, policymaker, journalist, citizen);
6. Producing a journalistic piece framing the topic for a wider audience;
7. A public exhibition or gallery with images, data, quotations, timelines, case evidence;
8. A piece of fiction, a movie, a book, a play, a podcast, a documentary…

D. Project Presentation

The final presentations will occur on **Dec 04** and **Dec 11** (groups will be randomly selected, unless a consensus is established among Team Leaders). In the final presentation, **each group will have 30 minutes** at most to present their project. This involves the following:

- Presenting the chosen case study, according to the guidelines in **Section B**.
- Presenting/conducting the practical activity, according to the guidelines in **Section C.**
- **The group has complete freedom on how to present the final project** (e.g., using a website, playing a video, producing a poster, demonstrating an activity, etc). It is not mandatory to deliver any final presentation (e.g., powerpoint).
- The only deliverable is the final report, submitted as a blog article, and the practical activity.

All groups should attend the final presentations.

Final Delivery

- By **Dec 20**, each group must submit in Moodle a link to the blog post describing the final project.

Case Studies

You can start by visiting the AI Incident Report Database, MIT Technology Review, The Markup "Show Your Work", "Machine Learning" , or "Pixel Hunt" walls, and selecting one case study of their interest. There are also interesting GitHub repositories that might provide some inspiration: Failed Machine Learning and Awful AI.

Important guidelines:

- The case study should be a **relevant societal concern related to AI development** and be **relevant to the topics discussed in the course** (data quality, bias/fairness, privacy, transparency, accountability, safety, autonomy, etc.).
- The case studies should **provide enough information for a comprehensive analysis**. This may include access to case details, used data, research papers, or reports (e.g., "one-shot" news in the media will be hard to map into a complete case study).